

End Semester Examination
M.Math II, Academic year: 2025-26
Rational Homotopy Theory
Date: 5/05/2026

Time: 3 Hours

Total Marks: 50

Instructions:

- Answer any five questions.
- Each question carries 10 marks.
- Clearly state all the results you use.

1. (a) Define **formal space**. (1 marks)
(b) Show that a retract of a formal space is formal. (2 marks)
(c) Show that the product of two formal spaces is formal. (2 marks)
(d) Let X be a well based CW complex. Show that the suspension ΣX is formal. (5 marks)
2. Let $(\Lambda V, d)$ be the minimal Sullivan model of $S^2 \vee S^3$.
(a) Determine the restriction of d to V^n for $n \leq 6$. (4 marks)
(b) Show that $S^2 \vee S^3$ is formal. (3 marks)
(c) Let $X = S^2 \vee S^2$. Show that $\pi_i(X)$ has a free part for all $i \geq 2$. (3 marks)
3. (a) Define the **triple Massey product**. What is its degree? (3 marks)
(b) Show that all triple Massey products vanish in a formal CDGA. (3 marks)
(c) For the DGA
 $(\Lambda(x, y, z), d), \quad dx = 0, \quad dy = 0, \quad dz = xy,$
show that there exists a non-trivial triple Massey product. (4 marks)
4. Describe the **bigraded (Halperin–Stasheff) model** for a cohomologically 1-connected graded algebra. (3 marks)

Let

$$H = \Lambda(x_1, x_2, x_3, x_4)/I,$$

where

$$|x_1| = |x_2| = |x_3| = 3, \quad |x_4| = 5,$$

and

$$I = \langle x_1x_2, x_1x_3x_4, x_2x_3x_4 \rangle.$$

Let

$$(\Lambda Z, d) \xrightarrow{\sim} (H, 0)$$

be its bigraded model.

- (a) Compute Z_0 .
 (b) Compute Z_1 .
 (c) Compute Z_2 . (7 marks)
5. (a) Compute the rational homotopy groups of $\mathbb{C}P^2$ and $\mathbb{C}P^3$. (3 marks)
 (b) Determine the minimal Sullivan model of $\mathbb{C}P^2 \times \mathbb{C}P^3$. (3 marks)
 (c) Write down $H^*(K(\mathbb{Q}, 3); \mathbb{Q})$ and compute $H^*(K(\mathbb{Q}, 4); \mathbb{Q})$ using spectral sequence. (4 marks)
6. (a) Consider the Sullivan representative of a map $f : (Y, *) \rightarrow (X, *)$ between simply connected spaces. Define the linear part $Q(f)$ of the Sullivan representative of f . (1 marks)
 (b) Define the pairing

$$(-; -) : V \times \pi_*(X) \rightarrow \mathbb{Q}.$$

 Show that it is bilinear. (3 marks)
 (c) Show that

$$(Q(f)v; \beta) = (v; \pi_*(f)(\beta)).$$

 (2 marks)
 (d) Define the Whitehead product. (1.5 marks)
 (e) Let $a \in \pi_{2n}(S^{2n})$ be the generator. Consider the Whitehead product $[a, a]_W \in \pi_{4n-1}(S^{2n})$. Prove that $[a, a]_W$ is not a torsion element in $\pi_{4n-1}(S^{2n})$. (2.5 marks)
7. (a) Define:
 • $\mathbb{Q}$ -local space
 • Rational homotopy equivalence
 • For a simply connected space X define $\mathbb{Q}$ -localization $(X)_{\mathbb{Q}}$ (3 marks)
 (b) Let G be a finitely generated abelian group. Compute $(K(G, n))_{\mathbb{Q}}$ for $n \geq 2$. (3 marks)
 (c) Show that for simply connected spaces X, Y :
 • $(X \times Y)_{\mathbb{Q}} = (X)_{\mathbb{Q}} \times (Y)_{\mathbb{Q}}$
 • $(X \vee Y)_{\mathbb{Q}} = (X)_{\mathbb{Q}} \vee (Y)_{\mathbb{Q}}$ (4 marks)

All the best!